

Notes: Golez & Jackwerth (JFE, 2012)

Pinning in the S&P 500 futures

Contents

1	Big picture	3
2	Data and measurement	3
2.1	Derivative markets in the paper	3
2.2	Main sample construction	4
2.3	Why serial SP expirations are the clean pinning laboratory	4
2.4	Pinning variables	4
2.5	Option activity variables	5
2.6	Futures volatility	5
3	Models, hypotheses, and empirical specifications	5
3.1	Testing whether pinning exists	5
3.2	Explaining pinning mechanisms	5
3.3	Mechanism 1: time decay of delta hedges	6
3.4	Mechanism 2: resale of slightly in-the-money options	6
3.5	Mechanism 3: early exercise of in-the-money options	6
3.6	Manipulation and volatility hypotheses	6
3.7	Cross-pinning from SPX options	7
4	Identification and empirical design	7
4.1	What identifies pinning	7
4.2	Why the serial SP sample helps	7
4.3	Why mechanism variables matter	8
4.4	Limits of the design	8
5	Tables and figures	9
5.1	Figures 1-3: hedging mechanisms	9
5.2	Figure 4 and Table 1: option activity and summary statistics	9
5.3	Figure 5 and Table 2: unconditional pinning and correlations	10
5.4	Table 3 and Figure 6: logistic pinning regressions and intraday timing	11
5.5	Table 4: cross-pinning from SPX options	12
5.6	Table A1 and Figure C1: institutional structure	13

6	Short summary	14
7	Conclusion	14
7.1	Core conclusion	14
7.2	Economic intuition	15
7.3	Takeaway	15

1 Big picture

- **Question.** Do S&P 500 futures prices get pulled toward option strike prices on option expiration days, and what trading mechanism creates this effect?
- **Core phenomenon.** *Pinning* means that the underlying futures settlement price finishes unusually close to the at-the-money option strike on expiration Friday.
- **Main setting.** The paper studies S&P 500 futures, serial options on S&P 500 futures (SP options), and options on the S&P 500 index (SPX options).
- **Key institutional detail.** Serial SP options expire monthly, but the underlying first-to-maturity futures continue trading for another month or two. This lets the futures price be pinned to the SP option strike.
- **Main empirical finding.** On serial SP option expiration days, first-to-maturity S&P 500 futures settle close to the nearest strike more often than expected under a locally uniform distribution.
- **Magnitude.** In the main long sample, futures settle within $\pm\$0.375$ of the strike 21.32% of the time, compared with about 15% under the uniform benchmark. In the more recent short sample, the rate is 22.47%.
- **Direction.** Pinning is stronger from below the strike than from above, consistent with larger short at-the-money put positions of market makers.
- **Mechanism.** Three forces interact: time decay of delta hedges, investor resale of in-the-money options, and early exercise of in-the-money options.
- **Important contrast with stock option pinning.** For individual stocks, market makers are often long options; for index options, market makers are more plausibly short. This reverses the implication of the Avellaneda-Lipkin time-decay mechanism from pinning to anti-pinning.
- **Net effect for SP serial options.** Time decay generates anti-pinning, but investor resale and early exercise generate enough pinning pressure to dominate.
- **Cross-market result.** SPX options on the cash index can induce *cross-anti-pinning* in futures because SPX options are cash settled and are often hedged with futures.
- **Manipulation result.** The evidence does not support manipulation as the main driver; futures volume does not explain pinning once hedging variables are included.
- **Economic takeaway.** Option expiration can temporarily move a very liquid index futures market. The authors estimate a notional futures value shift of at least \$115 million per expiration day, and about \$240 million in the later sample.

2 Data and measurement

2.1 Derivative markets in the paper

- **SP futures:** S&P 500 futures traded on the CME.
- **SP options:** options on S&P 500 futures, also traded on the CME. These are American options and can settle by physical delivery of futures.
- **SPX options:** options on the S&P 500 cash index, traded on the CBOE. These are cash settled and are European after April 1986.

Interpretation: The paper exploits the fact that these markets are closely linked but settle in different ways. Physical delivery for serial SP options creates resale and early-exercise channels. Cash settlement for SPX options makes the Avellaneda-Lipkin time-decay channel more dominant.

2.2 Main sample construction

- The futures data cover daily open, high, low, close, settlement prices, volume, and open interest from the introduction of S&P 500 futures in 1982 through 2009.
- SP option data come from the CME and include daily prices, volume, open interest, and early exercise.
- SPX option data come from Market Data Express and include daily prices, open interest, volume, OpenClose data, and special opening settlement values.
- The long serial SP option sample runs from November 1992 to November 2009.
- The short sample runs from October 1998 to November 2009 because early option exercise data are unreliable or mostly zero in the earlier period.
- The authors regularly exclude October 1987 and October 2008 crash months.

Interpretation: The long sample is chosen to avoid overlapping SP and SPX settlement regimes that would mix pinning and cross-pinning effects. The short sample is needed for the early-exercise variable.

2.3 Why serial SP expirations are the clean pinning laboratory

- Serial SP options expire in January, February, April, May, July, August, October, and November.
- The underlying first-to-maturity futures do not expire on those serial option expiration days.
- Therefore, option hedging trades can move the futures price without the futures price being mechanically tied to cash index settlement.
- Quarterly expirations are different because futures expire into the S&P 500 basket and hedging is more likely to use second-to-maturity futures.

Interpretation: The identification of pinning relies on expiration days when options expire but the underlying futures contract continues to trade.

2.4 Pinning variables

Let Fut_t be the first-to-maturity futures settlement price and let K_t^{ATM} be the strike nearest to that settlement price. The symmetric pinning indicator is

$$\text{Pinn_sym}_t = \begin{cases} 1, & \text{if } |Fut_t - K_t^{ATM}| \leq 0.375, \\ 0, & \text{otherwise.} \end{cases} \quad (1)$$

The paper also separates pinning by direction using a fair value of the first-to-maturity futures, denoted Fut_t^* , inferred from the second-to-maturity futures:

$$\text{Pinn_above}_t = \begin{cases} 1, & \text{Pinn_sym}_t = 1 \text{ and } Fut_t \leq Fut_t^*, \\ 0, & \text{otherwise,} \end{cases} \quad (2)$$

$$\text{Pinn_below}_t = \begin{cases} 1, & \text{Pinn_sym}_t = 1 \text{ and } Fut_t \geq Fut_t^*, \\ 0, & \text{otherwise.} \end{cases} \quad (3)$$

Interpretation: “Pinning from above” means the futures price is pushed downward toward the strike. “Pinning from below” means it is pushed upward toward the strike.

2.5 Option activity variables

- *OI*: at-the-money option open interest measured on the Thursday before expiration.
- *Call_OI* and *Put_OI*: call and put components of at-the-money open interest.
- *VOL*: at-the-money option trading volume on expiration Friday.
- *Call_VOL* and *Put_VOL*: call and put components of expiration-day volume.
- *OE*: at-the-money early option exercise on expiration Friday.
- *Call_OE* and *Put_OE*: call and put components of early exercise.
- *Fut_vol*: number of futures contracts traded on expiration Friday.

Interpretation: Open interest measures outstanding positions that must be delta hedged. Volume and early exercise measure closing or conversion of positions that can force market makers to rebalance hedges.

2.6 Futures volatility

Futures volatility is measured on the Thursday before expiration using the Parkinson scaled daily realized range:

$$Fut_sigma_t = \frac{[\log(Fut_high_t) - \log(Fut_low_t)]^2}{4\log(2)}. \quad (4)$$

Interpretation: The Thursday measure avoids using expiration-day volatility, which could itself be affected by pinning.

3 Models, hypotheses, and empirical specifications

3.1 Testing whether pinning exists

The baseline logistic regression is

$$Pr(\text{Pinn_sym}_t = 1) = \frac{1}{1 + \exp[-(\alpha + \beta \text{Dummy}_t)]}, \quad (5)$$

where $\text{Dummy}_t = 1$ on option expiration days and zero otherwise.

Interpretation: If $\beta > 0$, futures prices are more likely to finish near the strike on expiration days than on nearby non-expiration days.

3.2 Explaining pinning mechanisms

After establishing pinning, the authors estimate expiration-Friday regressions of the form

$$Pr(\text{Pinn}_t = 1) = \frac{1}{1 + \exp[-(\alpha + \gamma' X_t)]}, \quad (6)$$

where X_t contains option open interest, option volume, early exercise, futures volume, and futures volatility.

Interpretation: The second stage asks which hedging mechanism explains the presence and direction of pinning.

3.3 Mechanism 1: time decay of delta hedges

For a market maker short an at-the-money option, the delta hedge changes as expiration approaches. With short index-option positions, the Avellaneda-Lipkin mechanism predicts anti-pinning:

- high at-the-money open interest implies larger short option positions;
- as time passes, the hedge must be adjusted away from the strike;
- this weakens pinning and can create anti-pinning.

Hypothesis AL-1. At-the-money open interest decreases pinning.

Interpretation: In the regression, OI should have a negative coefficient if the time-decay hedge effect dominates.

3.4 Mechanism 2: resale of slightly in-the-money options

Individual investors dislike ending up with futures exposure after expiration. If their options are slightly in the money, they may sell them back to market makers before expiration.

- Selling in-the-money calls to market makers forces market makers to unwind hedges by selling futures, pushing the futures price downward toward the strike.
- Selling in-the-money puts forces the opposite hedge adjustment, pushing the futures price upward toward the strike.

Hypothesis AL-2. At-the-money option volume increases pinning.

Hypothesis AN-1. At-the-money call volume increases pinning from above.

Hypothesis AN-2. At-the-money put volume increases pinning from below.

Interpretation: Expiration-day volume is interpreted as evidence of investors closing or reselling positions, which induces market-maker hedge adjustments toward the strike.

3.5 Mechanism 3: early exercise of in-the-money options

Because SP options are American and settle by physical delivery, individual investors may early exercise in-the-money options before expiration.

- Early exercise of calls leads investors to sell delivered futures, pushing down toward the strike.
- Early exercise of puts leads investors to buy futures for delivery, pushing up toward the strike.

Hypothesis EARLY-1. At-the-money call early exercise increases pinning from above.

Hypothesis EARLY-2. At-the-money put early exercise increases pinning from below.

Interpretation: Early exercise has the same directional implication as resale, but the data are available only for the shorter sample.

3.6 Manipulation and volatility hypotheses

The paper also considers two alternatives:

- **Manipulation:** futures volume might capture deliberate attempts to move futures prices toward strikes.
- **Volatility:** high futures volatility can obscure or weaken pinning.

Hypothesis NI-1. Futures volume is insignificant after accounting for delta hedging.

Hypothesis NI-2. Futures volatility decreases pinning.

Interpretation: If pinning is mostly hedging driven, futures volume should not add explanatory power once option-activity variables are included.

3.7 Cross-pinning from SPX options

SPX options settle into the cash index and are hard to hedge directly with all 500 component stocks. Market makers may hedge SPX exposure with S&P 500 futures. This creates a cross-market test:

- SPX options can affect futures even though the options are written on the cash index.
- Because SPX options are cash settled, resale and physical-delivery channels are absent.
- Therefore the Avellaneda-Lipkin time-decay mechanism should dominate and create cross-anti-pinning.

Interpretation: Cross-pinning is a useful placebo-plus-mechanism test: the related market should show the opposite sign because the institutional settlement channel is different.

4 Identification and empirical design

4.1 What identifies pinning

- The main comparison is expiration Fridays versus nearby non-expiration days.
- The dependent variable is a local settlement event: whether the futures settlement price lies within $\pm\$0.375$ of the nearest strike.
- The uniform benchmark says that, with \$5 strike spacing, about 15% of local prices should fall in the \$0.75 pinning interval.
- The paper also uses logistic regressions with surrounding days to estimate whether expiration days are special.

Interpretation: If prices were locally smooth and unrelated to option strikes, there should be no spike at expiration strikes.

4.2 Why the serial SP sample helps

- Serial SP options expire while the first-to-maturity futures still trade.
- This avoids the mechanical settlement link that exists when futures themselves expire.
- It also makes option hedging pressure observable in the futures settlement price.

Interpretation: The design isolates option-induced price pressure rather than mechanical convergence of futures to the cash index.

4.3 Why mechanism variables matter

- Open interest, volume, and early exercise map directly to distinct hedging theories.
- Directional pinning from above or below helps distinguish call-driven and put-driven mechanisms.
- Futures volume is used as a check against manipulation-based explanations.
- Futures volatility is used as a control for market conditions that can mask pinning.

Interpretation: The paper is not content with showing a strike-price clustering fact; it asks which microstructure force generates the clustering.

4.4 Limits of the design

- The paper uses aggregate daily data rather than observing every market-maker hedge trade.
- SP option OpenClose data are not available, so market-maker positions are inferred from related SPX market evidence.
- Early exercise data are reliable only in the shorter late sample.
- The exact pinning interval is not theoretically pinned down, so robustness to alternative intervals is important.

Interpretation: The evidence is strongest when the directional predictions from institutional details, option variables, and cross-market tests all point in the same direction.

5 Tables and figures

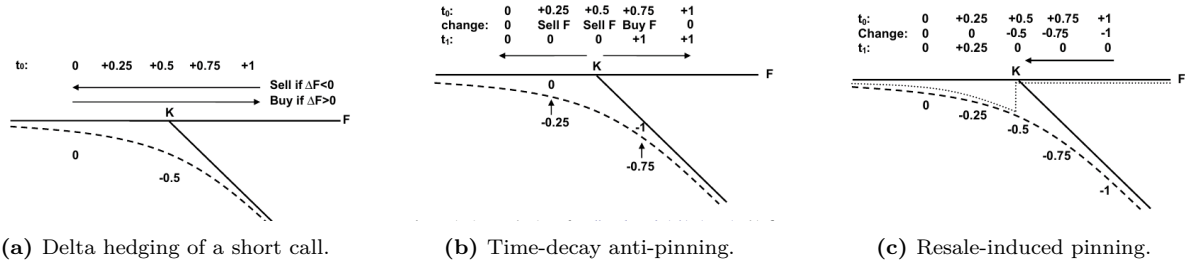
The visuals below are directly cropped from the paper and grouped compactly to speed reading.

5.1 Figures 1-3: hedging mechanisms

Read:

- Figure 1 shows how the delta hedge of a short call changes with the futures price near the strike.
- Figure 2 shows the Avellaneda-Lipkin time-decay mechanism: for short option positions, hedge adjustments can push the futures price away from the strike, generating anti-pinning.
- Figure 3 shows the Wall Street Journal resale mechanism: when investors sell in-the-money options back to market makers, market makers unwind hedges in a way that pushes the futures price toward the strike.

Economic message: The two mechanisms have opposite signs in this market. The empirical puzzle is why net pinning survives even though the time-decay mechanism predicts anti-pinning for short index-option positions.

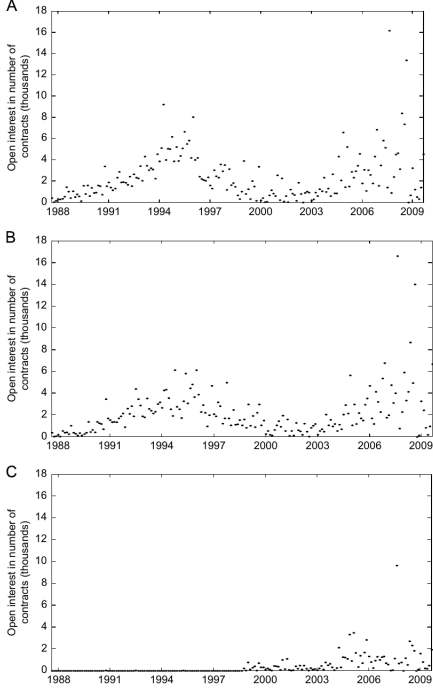


5.2 Figure 4 and Table 1: option activity and summary statistics

Read:

- Figure 4 shows rising at-the-money SP option activity over time: open interest, volume, and early exercise all become more important in the later years.
- Table 1 reports that symmetric pinning occurs 21.32% of the time in the long sample.
- Pinning from below is more common than pinning from above: 14.71% versus 6.62% in the long sample.
- Average at-the-money option volume is highly correlated with open interest and rises sharply in the later sample.

Economic message: The data show that pinning is not just a visual curiosity. It is more frequent than the 15% uniform benchmark and has a clear directional asymmetry.



(a) Time patterns of option activity.

Table 1

Summary statistics.

This table collects the summary statistics for the serial expiration dates in the period November 1992 to November 2009 (excluding October 2008). Symmetric pinning $Pinn_sym$ is a binary variable, which is one if the futures settlement price is within \$0.375 to the left or right of the at-the-money strike price and zero otherwise. $Pinn_above$ ($Pinn_below$) is a binary variable that takes a value of one if there is pinning and the futures settlement price is below (above) the fair value based on the second-to-maturity future and zero otherwise. At-the-money open interest OI is measured one day before the serial expiration day with respect to at-the-money strike price on the serial expiration day. At-the-money open interest is the sum of at-the-money call open interest $Call_OI$ and at-the-money put open interest Put_OI . At-the-money volume VOL and at-the-money option exercise OE are both measured on the serial expiration date with respect to the at-the-money strike price. At-the-money volume is the sum of at-the-money call volume $Call_VOL$ and at-the-money put volume Put_VOL . Similarly, at-the-money option exercise OE is the sum of at-the-money call option exercise $Call_OE$ and at-the-money put option exercise Put_OE . Futures volume Fut_vol measures number of futures contracts traded on the serial expiration Friday. Futures volatility Fut_sigma is a scaled realized daily range measured one day before the expiration date. We omit missing observations. Numbers in brackets next to number of observations denote the number of nonmissing observations for each variable. Summary statistics for OE are based on the period October 1998 to November 2009. Variables measuring options and futures trading activity (OI through Fut_vol) are expressed in thousands.

Variable	Number of observations	Mean	Standard deviation	Minimum	Maximum
$Pinn_sym$	136 (136)	0.2132	0.4111	0.0000	1.0000
$Pinn_above$	136 (136)	0.0662	0.2495	0.0000	1.0000
$Pinn_below$	136 (136)	0.1471	0.3555	0.0000	1.0000
OI	136 (129)	2.8363	2.4610	0.0240	16.1640
$Call_OI$	136 (130)	1.6522	1.5499	0.0020	8.0940
Put_OI	136 (134)	1.1771	1.5290	0.0010	11.6000
VOL	136 (135)	2.4753	2.2920	0.0840	16.5940
$Call_VOL$	136 (135)	1.4042	1.6679	0.0260	11.8200
Put_VOL	136 (136)	1.0633	1.1310	0.0020	5.8510
OE	89 (88)	0.8184	1.2123	0.0000	9.6610
$Call_OE$	89 (88)	0.4051	0.7288	0.0000	3.4800
Put_OE	89 (89)	0.4087	0.8877	0.0000	7.2720
Fut_vol	136 (136)	55.0479	23.9308	15.0600	160.5500
Fut_sigma	136 (136)	0.1384	0.0941	0.0357	0.6830

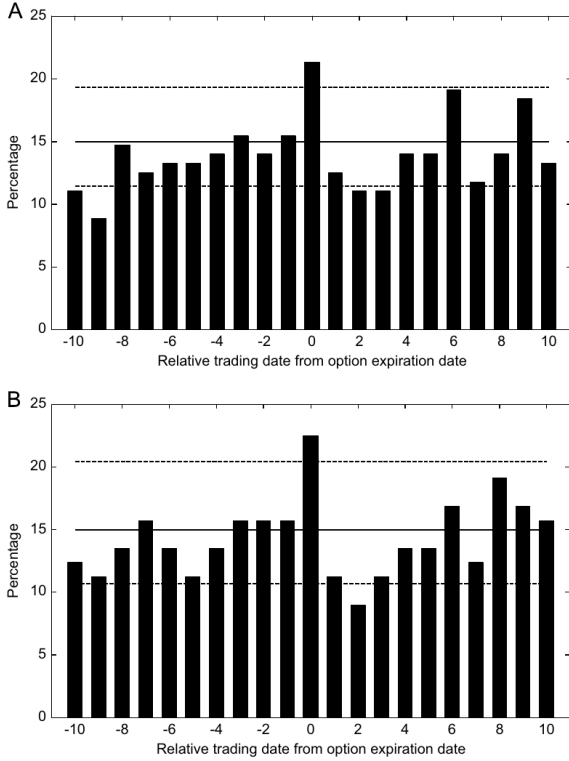
(b) Summary statistics.

5.3 Figure 5 and Table 2: unconditional pinning and correlations

Read:

- Figure 5 plots the percentage of futures settlements within the pinning band in the ten days before and after expiration.
- The expiration date itself shows the highest bar in both the long and short samples.
- Table 2 shows that option volume has the highest unconditional correlation with symmetric pinning, about 0.36.
- Open interest and early exercise have positive unconditional correlations with symmetric pinning, while futures volatility is negatively correlated.

Economic message: Option expiration is special, and option volume is the most visible raw correlate of pinning.



(a) Pinning percentages around expiration.

Table 2

Unconditional correlation structure for the main variables.

This table collects the unconditional correlations for the main variables for the serial expiration dates in the period November 1992 to November 2009 (excluding October 2008). Symmetric pinning *Pinn_sym* is a binary variable, which is one if the futures settlement price is within \$0.375 to the left or right of the at-the-money strike price and zero otherwise. At-the-money open interest *OI* is measured one day before the serial expiration day with respect to at-the-money strike price on the serial expiration day. At-the-money volume *VOL* and at-the-money option exercise *OE* are both measured on the serial expiration date with respect to the at-the-money strike price. Futures volume *Fut_vol* measures number of futures contracts traded on the serial expiration Friday. Futures volatility *Fut_sigma* is a scaled realized daily range measured one day before the expiration date. We omit missing observations. Correlations for *OE* are based on the period October 1998 to November 2009.

Variable	<i>Pinn_sym</i>	<i>OI</i>	<i>VOL</i>	<i>OE</i>	<i>Fut_vol</i>	<i>Fut_sigma</i>
<i>Pinn_sym</i>	1.0000	0.2223	0.3641	0.2235	-0.1132	-0.1264
<i>OI</i>		1.0000	0.8845	0.7903	-0.1057	-0.1126
<i>VOL</i>			1.0000	0.7546	-0.1180	-0.0941
<i>OE</i>				1.0000	-0.1450	-0.0361
<i>Fut_vol</i>					1.0000	0.1581
<i>Fut_sigma</i>						1.0000

(b) Unconditional correlations.

5.4 Table 3 and Figure 6: logistic pinning regressions and intraday timing

Read:

- Table 3, Model 1, finds strong evidence of symmetric pinning in the long sample, with a p -value of 0.01.
- Model 2 confirms the result in the short sample, with a p -value of 0.03.
- Models 3 and 4 show that pinning is asymmetric: pinning from below is strong, while pinning from above is not significant in the basic directional test.
- In mechanism regressions, open interest tends to reduce pinning, consistent with the Avellaneda-Lipkin anti-pinning channel.
- Option volume increases pinning, consistent with the resale mechanism.
- Futures volume does not explain pinning after the option hedging variables are included.
- Figure 6 shows that pinning becomes especially pronounced in the final part of the trading day, beginning around 14:45.

Economic message: The regression evidence supports a hedging interpretation. Time decay pushes away from the strike, but expiration-day option trading pushes back toward the strike and dominates in the serial SP sample.

Table 3

Logistic regressions for pinning.

This table presents results of the logistic regressions showing pinning, based on our long sample (November 1992 to November 2009; superscript L) and our short sample (October 1998 to November 2009; superscript S). Symmetric pinning $Pinn_sym$ is a binary variable, which is one if the futures settlement price is within \$0.375 below or above the at-the-money strike price and zero otherwise. $Pinn_above$ ($Pinn_below$) is a binary variable that takes a value of one if there is pinning and the futures settlement price is below (above) the fair value based on the second-to-maturity future and zero otherwise. At-the-money open interest OI is measured one day before the serial expiration day with respect to at-the-money strike price on the serial expiration day. At-the-money open interest is the sum of at-the-money call open interest $Call_OI$ and at-the-money put open interest Put_OI . At-the-money volume VOL and at-the-money option exercise OE are both measured on the serial expiration date with respect to the at-the-money strike price. At-the-money volume is the sum of at-the-money call volume $Call_VOL$ and at-the-money put volume Put_VOL . Similarly, at-the-money option exercise OE is the sum of at-the-money call option exercise $Call_OE$ and at-the-money put option exercise Put_OE . Futures volume Fut_vol measures number of futures contracts traded on the serial expiration Friday. Futures volatility Fut_sigma is a scaled realized daily range measured one day before the expiration date. We omit missing observations. Variables measuring options and futures trading activity (OI through Fut_vol) are expressed in thousands. We report p -values in parentheses below each coefficient. Finally, we report the number of observations used for each regression.

Right-hand side regression variables	1 $Pinn_L^{sym}$	2 $Pinn_L^{above}$	3 $Pinn_L^{below}$	4 $Pinn_L^{sym}$	5 $Pinn_L^{above}$	6 $Pinn_L^{below}$	7 $Pinn_L^{sym}$	8 $Pinn_L^{above}$	9 $Pinn_L^{below}$	10 $Pinn_L^{sym}$	11 $Pinn_L^{above}$	12 $Pinn_L^{below}$	13 $Pinn_L^{sym}$	14 $Pinn_L^{above}$
Constant α	-1.85 (0.00)	-1.82 (0.00)	-2.64 (0.00)	-2.58 (0.00)	-2.15 (0.00)	-2.01 (0.00)	-2.02 (0.00)	-2.62 (0.00)	-2.43 (0.00)	-2.67 (0.00)	-2.38 (0.00)	-2.09 (0.02)	-0.27 (0.75)	-1.73 (0.00)
Dummy _t	0.54 (0.01)	0.59 (0.03)	-0.01 (0.99)	0.83 (0.00)										-0.22 (0.42)
Open Interest OI_{t-1}					-0.35 (0.09)	-0.86 (0.02)	-0.87 (0.02)						-0.87 (0.02)	-1.08 (0.01)
Volume VOL_t					0.69 (0.00)	1.13 (0.00)	1.13 (0.00)						1.13 (0.00)	1.28 (0.00)
Option Exercise OE_t							0.05 (0.92)						0.05 (0.92)	0.13 (0.75)
$Call_OI_{t-1}$								-1.07 (0.04)			-1.33 (0.06)			
$Call_VOL_t$								0.99 (0.03)			1.06 (0.07)			
$Call_OE_t$											0.49 (0.66)			
Put_OI_{t-1}									-0.48 (0.11)		-0.88 (0.22)			
Put_VOL_t									1.01 (0.00)		1.40 (0.02)			
Put_OE_t											0.10 (0.91)			
Fut_vol_t												0.00 (0.93)		
Fut_sigma_{t-1}													-13.03 (0.03)	
Number of observations	2813	1839	2813	2813	129	82	82	130	134	83	87	82	82	2813

Table 3: Logistic regressions for pinning.

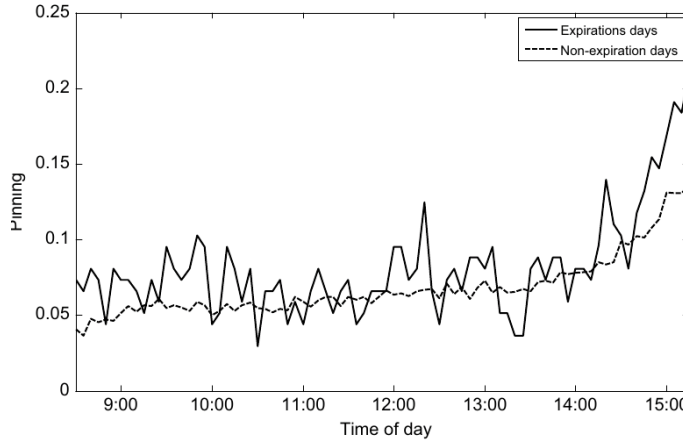


Figure 6: Intraday pinning rises sharply late in the trading day.

5.5 Table 4: cross-pinning from SPX options

Read:

- Table 4 tests whether SPX options on the cash index create pinning or anti-pinning in S&P 500 futures.
- Model 15 shows cross-anti-pinning: futures are less likely to finish near the adjusted SPX strike on SPX expiration days.
- This sign is intuitive because SPX options are cash settled, so physical-delivery resale and early exercise effects are absent.

- The remaining dominant channel is time decay of delta hedges, which predicts anti-pinning when market makers are short index options.

Economic message: The cross-pinning test strengthens the mechanism story. Closely related index options affect futures prices, but the sign changes when the settlement institution changes.

Table 4
Logistic regressions for cross-pinning.

This table presents results of the logistic regressions showing cross-pinning. Symmetric pinning $Pinn_sym$ is a binary variable, which is one if the futures settlement price is within \$0.375 below or above the at-the-money strike price (appropriately adjusted for the cost-of-carry) and zero otherwise. At-the-money open interest OI is measured on the same day as cross-pinning with respect to at-the-money strike price of the SPX options. At-the-money volume VOL and at-the-money close buy of small and medium customers CB are both measured on the same day as cross-pinning with respect to the at-the-money strike price of the SPX options. Futures volume Fut_vol measures number of futures contracts traded. Futures volatility Fut_sigma is a scaled realized daily range measured one day before the day on which we test for cross-pinning. We omit missing observations. Variables measuring options and futures trading activity (OI through Fut_vol) are expressed in thousands. We report P -values in parentheses below each coefficient. Finally, we report the number of observations for each regression. Model 15 is based on the 245 expiration days in which we check for cross-pinning (see Appendix C for details) against the surrounding ± 10 day, and Models 16 to 20 are based on the 122 day in which we test for cross-pinning from AM expiring SPX options.

Right-hand side Regression variables	15 $Pinn_sym_t$	16 $Pinn_sym_t$	17 $Pinn_sym_t$	18 $Pinn_sym_t$	19 $Pinn_sym_t$	20 $Pinn_sym_t$	21 $Pinn_sym_t$	22 $Pinn_sym_t$	23 $Pinn_sym_t$	24 $Pinn_sym_t$	25 $Pinn_sym_t$
Constant α	-1.76 (0.00)	-1.53 (0.01)	-1.34 (0.01)	-1.44 (0.02)	-1.32 (0.01)	-1.51 (0.01)	-1.39 (0.01)	-0.53 (0.60)	-0.90 (0.29)	-0.66 (0.50)	-0.63 (0.50)
Dummy _t	-0.51 (0.02)										
Open Interest OI_t		-0.03 (0.12)		-0.02 (0.39)		-0.03 (0.13)		-0.02 (0.26)		-0.03 (0.12)	
Volume VOL_t				-0.02 (0.67)							
Close Buy CB_t						-0.05 (0.94)		-0.32 (0.64)		-0.08 (0.90)	
Put_ OI_t			-0.11 (0.05)		-0.11 (0.12)		-0.12 (0.05)		-0.11 (0.08)		-0.13 (0.05)
Put_ VOL_t					-0.01 (0.93)						
Put_ CB_t							0.59 (0.61)		0.58 (0.63)		0.77 (0.53)
Fut_ vol_t								-0.02 (0.25)	-0.01 (0.48)		
Fut_ $sigma_{t-1}$										-7.72 (0.29)	-7.66 (0.34)
Number of observations	4440	122	122	122	122	122	122	122	122	122	122

Table 4: Logistic regressions for cross-pinning.

5.6 Table A1 and Figure C1: institutional structure

Read:

- Table A1 summarizes the main characteristics of SPX options, SP futures, and SP options.
- SP serial options are physically settled into futures, whereas SPX options are cash settled.
- Figure C1 lays out the sample structure across time, settlement regimes, and product types.
- The sample design is built around avoiding periods in which SP and SPX options create overlapping forces that are hard to disentangle.

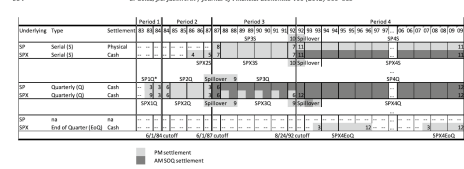
Economic message: The institutional details are not background trivia. They are the reason the same hedging logic can produce pinning in one market and anti-cross-pinning in another.

Table A1

Main characteristics of the Standard & Poor's (S&P) 500 derivatives.

This table collects the main characteristics of the S&P 500 index related derivatives. Q stands for quarterly expiration cycle (March, June, September, and December), S stands for serial expiration months (January, February, April, May, July, August, October, and November). AM settlement price is based on the special opening value of the underlying, PM settlement price is based on last prices of the underlying on an expiration day. The table is based on the information obtained from the Chicago Mercantile Exchange (CME) webpage (<http://www.cmegroup.com>), Chicago Board of Options Exchange (CBOE) webpage (www.cboe.com), Stoll and Whaley (1986, 1991), and CBOE Regulatory Circular Number RC92-46. * indicates that serial expirations for SPX options were introduced in April 1986.

Characteristic	SPX options	SP futures	SP options
Underlying	S&P 500 index	S&P 500 index	SP futures
Opening date	July 1, 1983	April 21, 1982	January 28, 1983
Strike price interval	5 points (first nearest month)	–	5 points (first and second nearest month) 10 and 25 points (deferred months)
Type	25 points (deferred months) European (American before April 1986)	–	American
Trading hours (central time)	8:30 AM–3:15 PM	8:30 AM–3:15 PM	8:30 AM–3:15 PM
Expiration months	Three serial months Three months in the quarterly cycle	Eight months in the quarterly cycle	Three serial months Eight months in the quarterly cycle
Settlement at expiration	Q+S: cash settlement	Q: cash settlement	Q: cash settlement S: physical delivery
Until June 1984	Q: PM settlement (third Friday)	Q: PM settlement (third Thursday)	Q: PM settlement (third Thursday)
June 1984 to June 1987	Q+S*: PM settlement (third Friday)	Q: PM settlement (third Friday)	Q: PM settlement (third Friday)
June 1987 to December 1993	Q+S: coexistence of AM and PM. settled options (third Friday)	Q: AM settlement (third Friday)	Q: AM settlement (third Friday)
December 1993 onward	Q+S: AM settlement (third Friday)	Q: AM settlement (third Friday)	S: PM settlement (third Friday) Q: AM settlement (third Friday) S: PM settlement (third Friday)

(a) Derivative market characteristics.**Fig. C1. Sample structure**
This table details the sample structure of the data used in the article. Underlying: Underlying Time when the Underlying is a SP or S&P 500. The**(b) Sample structure.**

6 Short summary

- The paper studies whether S&P 500 futures are pinned to option strikes on expiration days.
- The main pinning sample uses serial SP options because the options expire while the underlying futures continue trading.
- Symmetric pinning is defined as settlement within $\pm \$0.375$ of the nearest strike.
- Futures settle close to the nearest strike more often on serial SP expiration days than on non-expiration days.
- In the long sample, the pinning rate is 21.32%, above the 15% uniform benchmark.
- Pinning from below is stronger than pinning from above.
- Open interest is negatively related to pinning, consistent with time-decay anti-pinning for short market-maker option positions.
- Option volume is positively related to pinning, consistent with resale of in-the-money options by individual investors.
- Early exercise has the expected directional interpretation but is harder to establish statistically because of the short sample.
- Futures volume is not a strong explanation after hedging variables are included, weakening a manipulation interpretation.
- SPX options create cross-anti-pinning in futures, consistent with the absence of physical-delivery channels.
- The notional futures value affected is economically meaningful: at least \$115 million per expiration day and about \$240 million in the later sample.

7 Conclusion

7.1 Core conclusion

Golez and Jackwerth show that S&P 500 futures exhibit pinning on serial SP option expiration days. This effect is not simply stock-option pinning scaled up to an index market. The institutional structure of the

derivative contracts matters, especially whether options are cash settled or physically delivered and whether the underlying futures continue to trade.

7.2 Economic intuition

The key intuition is that option-market hedging can temporarily move even highly liquid futures prices. Market makers who are short options must rebalance hedges as option deltas change. Time decay pushes futures away from strikes, but investor resale and early exercise of in-the-money SP options force hedge unwinding that pushes futures toward strikes. In the serial SP option market, the toward-strike forces dominate.

7.3 Takeaway

Pinning is an expiration-day microstructure effect driven by derivative hedging. The paper's most useful lesson is that the sign of the effect is not universal. The same delta-hedging logic can generate pinning or anti-pinning depending on who holds the options, how the options settle, and which market is used for hedging.